MILITARY SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM 24 VOLTS, 20 AMPERE-HOUR (LOW FREQUENCY VIBRATION)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the battery assembly described herein shall consist of this document and the latest issue of specification MIL-B-81757.

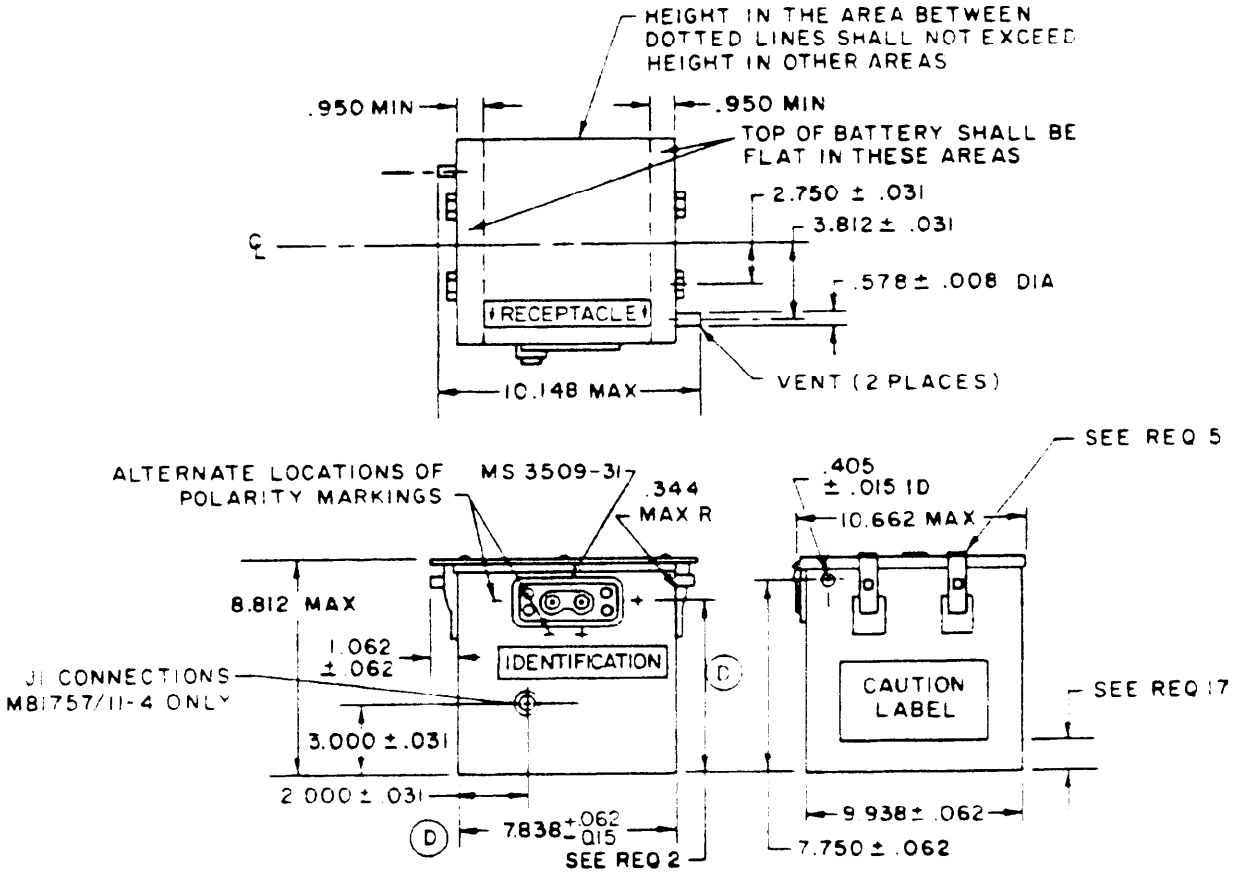


FIGURE 1. Battery assembly.
P/N M81757/11-3 and M81757/11-4

AMSC N/A

FSC 6140

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:07Z
Check the source to verify that this is the current version before use.

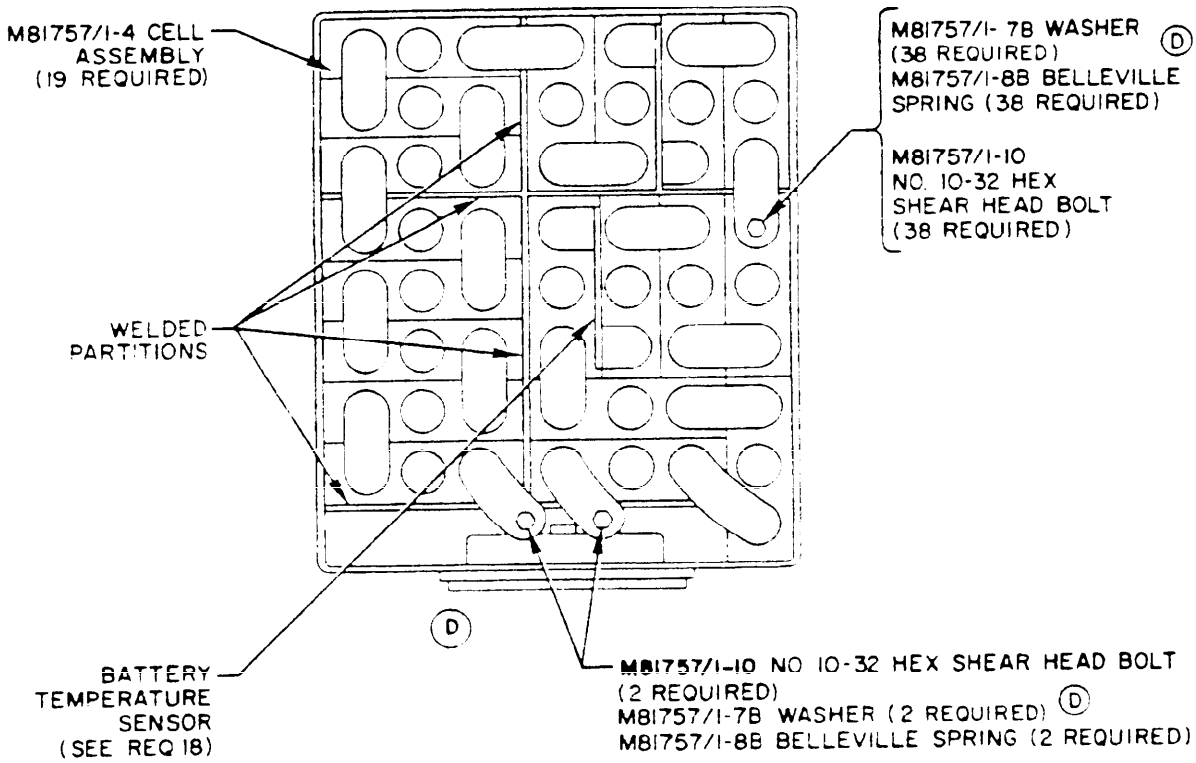


FIGURE 2. Cell arrangement and strap assignment.
P/N M81757/11-3 and M81757/11-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:07Z
Check the source to verify that this is the current version before use.

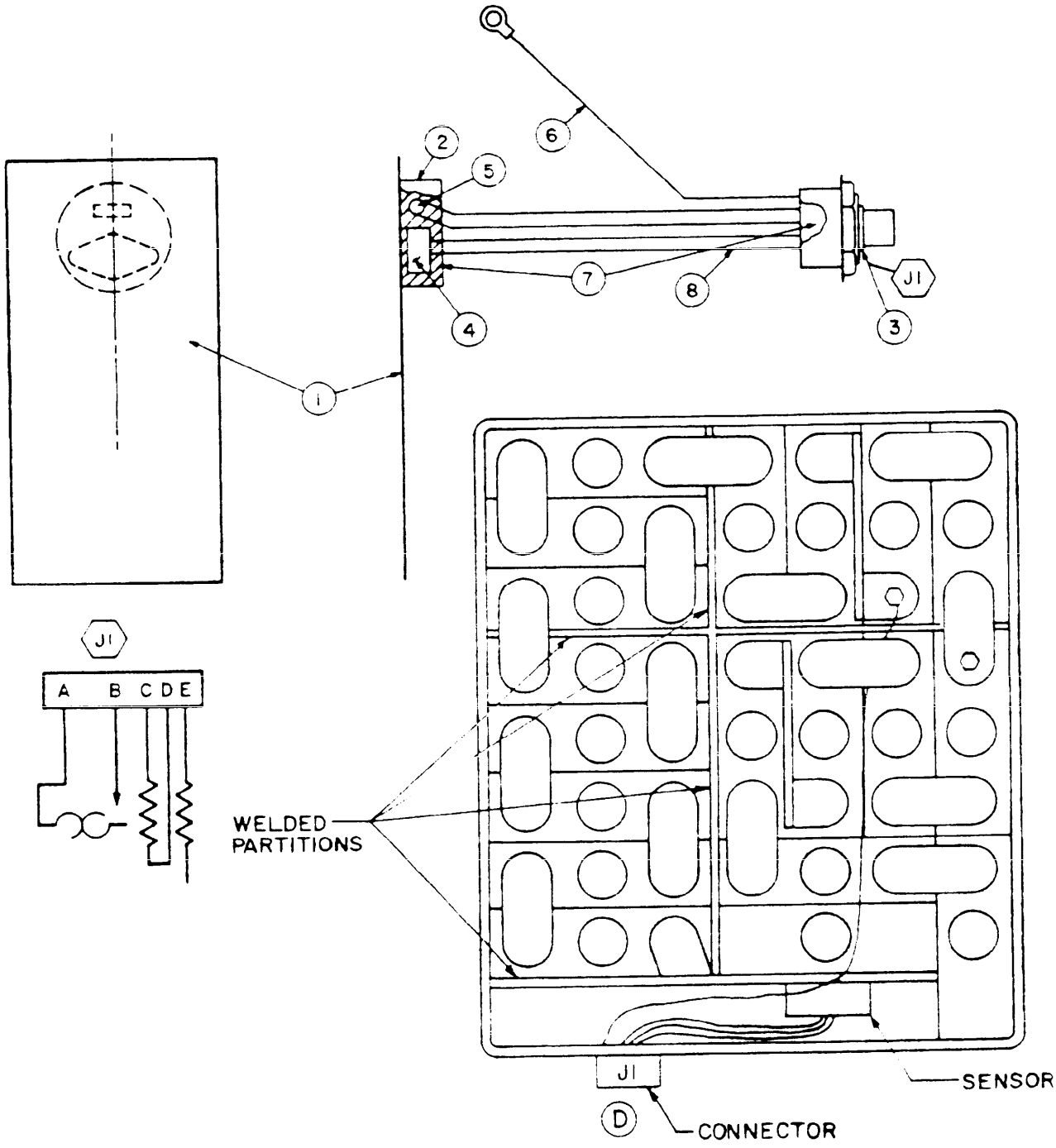


FIGURE 3. Cell arrangement and battery sensor/connector assembly.
(Battery assembly P/N M81757/11-4 only)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:07Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

1. Battery M81757/11-4 shall include a battery sensor/connector assembly capable of withstanding all environmental conditions to which the battery is subjected without damage or degradation of performance. The battery sensor plate shall be in thermal contact with the front surface of cell 19. No air flow shall be permitted between the surfaces of cell 19 and the battery sensor plate. The battery sensor/connector assembly shall consist of:

   a. One temperature sensor plate (1) with component mounting cavity (2), installed into the battery as shown.

   b. One auxiliary receptacle (3), having ten contacts, type D38999/44WC98SN, conforming to MIL-C-38999, shall be mounted to the battery container as shown.

   c. One thermal switch (4), in accordance with M24236/1-1565, having normally open contacts, which close at 71° ± 2.8°C and reopen at 62.8 ± 2.8°C. The thermal switch shall be certified to withstand a minimum of 50,000 operating cycles at 30 VDC and 2.0 amperes without the switch's opening and closing temperature values changing more than ± 2.8°C. The thermal switch shall be potted as shown, with the switch leads connected to pins A and B of the battery auxiliary connector.

   d. One 1/4-watt wire-wound resistor (5), in accordance with MIL-R-39005, valued at 3,000 ohms ± 1 percent at 25°C, and shall have temperature coefficients of:

      T.C. = 3,900 ± 300 PPM/°C below 25°C
      T.C. + 4,100 ± 300 PPM/°C above 25°C

   The resistor shall be potted as shown, with the resistor leads connected to pins C and D of the battery auxiliary receptacle.

   e. One center tap voltage lead (6), having a nickel-plated terminal lug, shall be connected to the negative terminal of cell 11 (referenced from battery negative). The center tap voltage lead shall include a 2,000 ohm, 1/8-watt resistor, in accordance with MIL-R-39017, and be connected to pin E of the battery auxiliary connector.

   f. All electrical connections made to the battery auxiliary connector and battery sensor assembly shall be potted (7) with a compound that will not support flame, absorb moisture, or be damaged by battery electrolyte.

   g. All wire (8) used for electrical connections shall be #20 AWG, in accordance with MIL-W-16878, teflon insulated, and capable of withstanding the temperature and exposure to battery electrolyte without damage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:07Z
Check the source to verify that this is the current version before use.

2. The receptacle terminal block shall be 0.866 ± 0.015 from the centerline of the contact pins to the top surface of the terminal block. The receptacle will be so located that the top of the terminal block and the cell terminal contact surfaces are in the same plane. Receptacle shall conform to MS3509-31.

(D) 3. All dimensions are prior to epoxy coating except: Overall width - 10.148 maximum, overall length - 10.662 maximum, overall height - 8.812 maximum, overall base width - 7.838 +.062 -0.15, and overall base length 9.938 ±.062.

4. The cell filler cap shall be in accordance with MIL-B-81757/2.

5. Spacers shall be polyethylene sheet in accordance with ASTM STD D4066-82, Composition A, Type I and Type III, minimum 0.015 inch thick, color natural.

6. Latches and strikes shall not be epoxy coated.

7. The battery voltage shall be 24 volts (nominal).

8. The battery weight shall be 53.5 ± 1.5 pounds when filled.

9. The battery shall contain 19 cells arranged as shown in Figure 2.

10. The rated capacity at 1C rate shall be 20 A.H./1 Hr/80°F/18.0V.

11. Insulating cover liner. An epoxy glass liner .062 ± .007 inch thick shall be contained between the inside, top surface of the metal container cover and the elastomer liner. The liner shall provide complete coverage of the inside, top surface of the container cover.

12. Hold-down pads. The elastomer liner shall include hold-down pads that contact the intercell connecting strap of each cell to restrict vertical movement of the cells during vibration.

13. Mounting for vibration testing. The battery shall be clamped in the cradle by two flat bars, with one of the bars bearing on the cover adjacent to each of the left and right edges of the cover, and extending inward from the edges 0.844 ± 0.100 from the sides, measured in the lower 2/10 of the height, and extending from the front edge of the cover to the back edge of the cover. The hold-down bars shall be clamped to the cradle by four vertical 5/16-24 threaded rods with nuts tightened to 45 ± 3 inch-pounds, bearing on the hold-down bars through cross-beams. The rods shall be arranged symmetrically about the battery, with two rods on each side of the battery. Two of the rods shall be 2.500 ± 0.100 inches from the front of the battery, and two of the rods shall be 2.500 ± 0.100 inches from the back of the battery, measured in the lower 2/10 of the height of the battery.

14. All cells within a battery shall be from the same production lot.

15. Cell and receptacle removal shall be impeded by non-removable fasteners (M81757/1-10).

Source: https://assist.dla.mil -- Downloaded: 202610-23T06:07Z
Check the source to verify that this is the current version before use.

6. The battery shall conform to MIL-B-81757 except as modified by the following:

3.5.7 Cells. Delete.

3.5.7.1 Mountings. Delete.

3.6.5.2.1 Cell terminal marking. Delete last sentence.

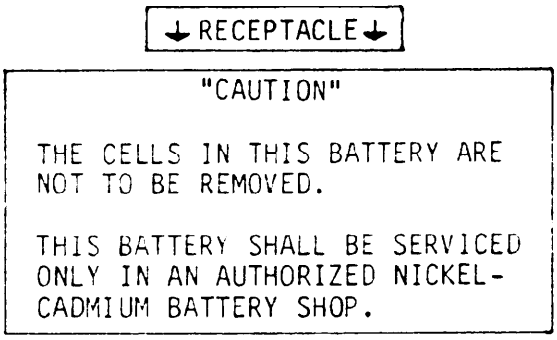
3.6.5.2.2 Cell identification. Delete.

3.6.5.2.3 Battery identification. Each battery shall have its identification marking silk screened with an epoxy overlay, or equal, that is impervious to 1.3 ± 0.025 specific gravity KOH electrolyte, in accordance with applicable specification sheet. Marking shall be in white print in accordance with MIL-M-13231. The marking shall show the following:

Information Required	Example
Battery, Storage	Battery, Storage
Military Battery Part No.	M81757/11-2
Replaces (include replacement data from applicable battery specification sheet)	Replaces MS90377-1
National Stock Number	NSN 6140-01-***-****
Voltage (Nominal)	24 Volts
Maximum Weight	Maximum Weight 55 Lbs
Military "C" Rated Capacity	(20 A.H./1 Hr/80°F/18.0V)
Contract Number	***
Date Code (see 3.6.5.2.5)	Date Code ***
Manufacturer's Name	***
Manufacturer's Part Number	***
Manufacturer's Serial Number	***
Warranty Expiration Date	041787

3.6.5.2.4 Battery caution and receptacle markings. These markings, in accordance with applicable specification sheet, shall be in white printing in accordance with MIL-M-13231. They shall be impervious to 1.3 ± 0.025 specific gravity KOH electrolyte. They shall be located as shown on the applicable specification sheet and read as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:07Z
Check the source to verify that this is the current version before use.



3.6.18 Slope discharge. During two slope discharge profiles, shown in Figure 3 the battery potential shall not be less than 12 volts.

3.6.19 Low frequency vibration. The battery shall meet the specified minimum requirement of 12 volts during the slope discharge profile shown in Figure 3 for each of the 17 cycles described in paragraph 4.6.25. The battery shall not exhibit an increase in charge current of more than 5 amperes from the lowest value reached. There shall be no physical damage to cells or hardware.

TABLE V. Qualification inspection of batteries. Add the following:

Test No.	Examination and Tests	Sample Number 1	2	3	4	Req'mt Para.	Method of Insp. Para.
20	Low Frequency Vibration	X				3.6.19	4.6.25

TABLE IX. Group C inspection. Add the following after internal pressure test:

Examination and Tests	Requirement Paragraph	Method of Inspection Paragraph
Subgroup I		
Slope Discharge	3.6.18	4.6.24

4.6.24 Slope discharge. The battery shall be subjected to the slope discharge (Figure 3 at 37.8° ± 2.8°C (100° ± 5°F)), immediately followed by a 5-hour 1.460 ± .010 volt/cell constant potential charge at room ambient temperature. Upon completion of the constant-potential charge, the battery shall be stabilized at 37.8° ± 2.8°C (100° ± 5°F), and within 12 hours subjected to the slope discharge (Figure 3). The battery shall meet the requirements of 3.6.18.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:07Z
Check the source to verify that this is the current version before use.

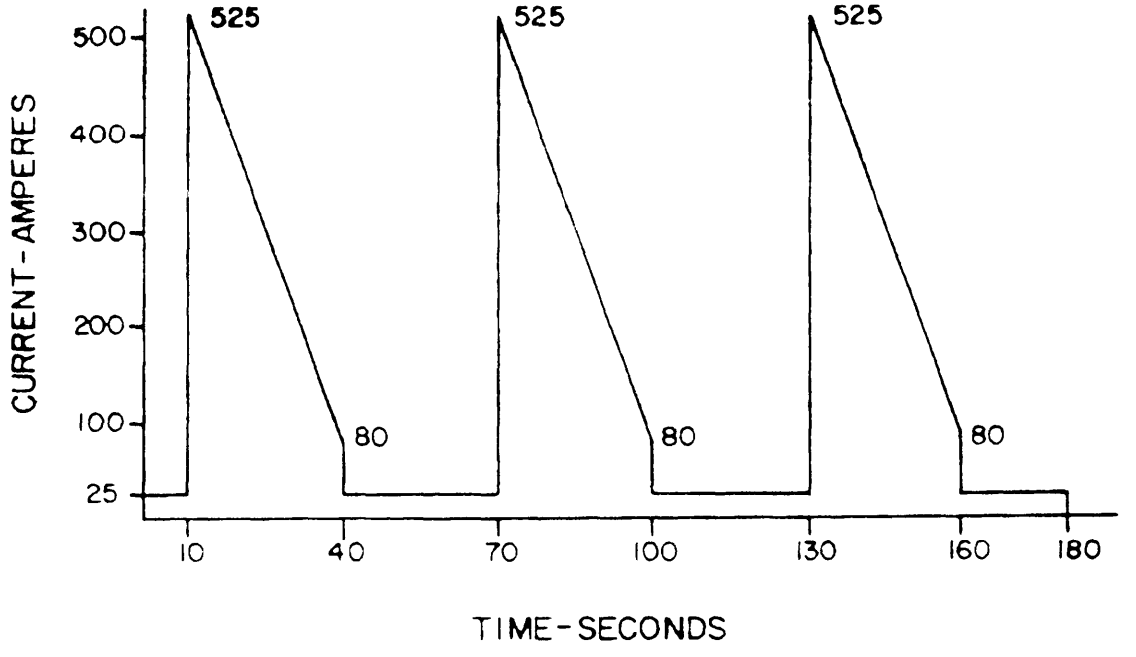


FIGURE 4. Slope discharge profile.

4.6.25 Low frequency vibration. The battery shall be subjected to 17 5-hour cycles of vibration along its vertical axis. The battery shall be mounted on a vibration fixture and the hold-down bars shall be clamped to the fixture by four vertical threaded rods with nuts tightened to 45 ± 3 inch-pounds. The battery shall not be subjected to more than two 5-hour cycles in any 24-hour period. The battery temperature at the start of each cycle shall be 37.8° ± 2.8°C (100° ± 5°F). A sensor (thermocouple or thermistor) shall be used for the purpose of monitoring battery temperature during testing. The sensor shall be positioned between cells 15 and 16 (referenced from battery negative) in the approximate center of the cell container's widest outer surface (equidistant from the top and bottom and from side to side). Prior to each 5-hour vibration period, the battery shall be subjected to the slope discharge (Figure 3). There shall be no vibration during the discharge. Upon completion of the discharge, the battery shall be removed from the 37.8° ± 2.8°C (100° ± 5°F) environment and immediately vibrated in room ambient temperature for a period of 5 hours. During the first 1.5 minutes, the acceleration level shall be 11g at 17 Hz the next 6 minutes, 5g at 17 Hz and the next 12.5 minutes 2g at 17 Hz. The g levels and times shall be repeated continuously throughout the 5-hour vibration period. At the start of the vibration and throughout the 5-hour period, the battery shall be subjected to a constant-potential charge equivalent to 1.460 ± .010 volt cell. Upon completion of each 5-hour vibration and charge, the battery temperature shall be stabilized at 37.8° ± 2.8°C (100° ± 5°F) prior to start of next cycle. Should the battery for any reason remain in open circuit condition for a period in excess of 24 hours, it shall be boost charged to a full-charge condition prior to resumption of testing. Maintenance of the battery shall not be permitted during the 17 cycles. The battery shall meet the requirements of 3.6.19.

Source: https://assist.dla.mil -- Downloaded: 202610-23T06:07Z
Check the source to verify that this is the current version before use.

5.5 Marking. The individual battery package shall be marked on one of the two largest vertical surfaces with the supersession data from note 2. The lettering shall be at least 1/2-inch high on battery package. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

17. The base dimensions of the container shall be within tolerance for a distance of 1.8 inches up from the bottom.

18. A sensor (thermocouple or thermistor), when required, shall be located as shown for the purpose of monitoring battery temperature during testing to this specification. The sensor shall be positioned in the approximate center of the cell container's widest outer surface (equidistant from the top and bottom, and from side to side).

19. The sides and bottom of the battery container shall be cold rolled, low carbon steel with a minimum 16 gauge thickness. The front and back panel of the container shall be cold rolled, low carbon steel with a minimum 20 gauge thickness. The container cover shall be stainless steel with a minimum 18 gauge thickness.

(D) 20. Method of mounting latches and strikes shall be optional. They shall conform to Corbin Cabinet Lock Division, American Hardware, item number 15834, or Nielson Latch number C-833314-2-SS-302 or equivalent.

NOTES:

1. All dimensions are in inches, unless otherwise specified.

2. Supersession data: M81757/11-3 and M81757/11-4 supersede MS90377, MS90321, and M81757 part numbers as follows when a product has been qualified for M81757/11-3 and M81757/11-4 part numbers.

Part Number	Replaces Part Numbers
M81757/11-3	MS90377-1, MS90321-79W, M81757/11-1
M81757/11-4	MS90377-1, MS90321-79W, M81757/11-2

Part Number	Superseded By
MS90377-1, MS90321-79W, M81757/11-1	M81757/11-3
MS90377-1, MS90321-79W, M81757/11-2	M81757/11-4

Preparing activity:
Navy - AS
(Project No. 6140-N643)

U.S. GOVERNMENT PRINTING OFFICE: 1987 - 704-054/72478

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:07Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Systems Engineering and Standardization Department
(SESD), Code 93
Lakehurst, NJ 08733-5100

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Systems Engineering and Standardization Department
(SESD), Code 93
Lakehurst, NJ 08733-5100

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:07Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-81757/11D(AS)	BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM 24 VOLTS, 20 AMPERE-HOUR (LOW FREQUENCY VIBRATION)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-23T06:07Z
Check the source to verify that this is the current version before use.